AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2024

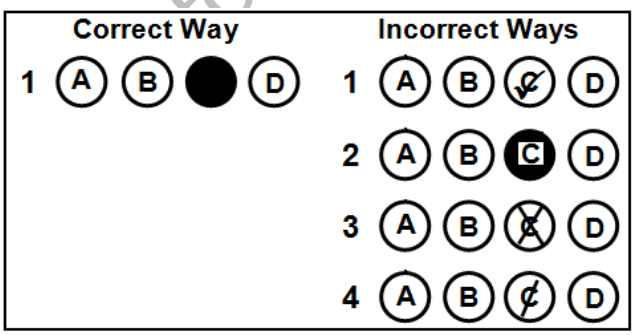
Physics Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

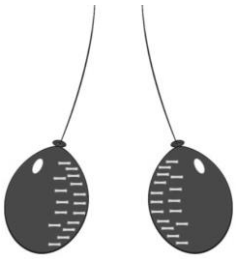
1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. The given diagram shows two like-charged balloons **Y** and **Z** that repel each other.



Which of the following factor(s) will affect the repulsive force between the balloons?

I. Sizes of the balloons
II. Distance between the balloons
III. Amount of charges on each balloon

A. I only
B. III only
C. I and II
D. II and III

2. During a charging of a capacitor in an electrical circuit, the charges

A. increase linearly.
B. decrease linearly.
C. increase exponentially.
D. decrease exponentially.

3. The purpose of the presence of a dielectric between the two plates of a capacitor is to

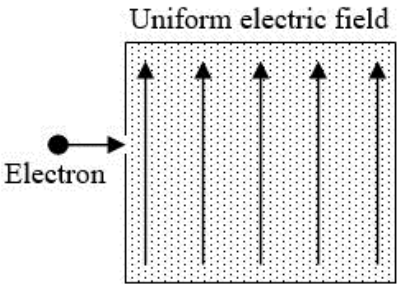
A. increase the capacitance.
B. decrease the capacitance.
C. make the capacitance zero.
D. make the capacitance constant.

4. The ink-jet printer and photocopier work on the principle of

A. electricity.
B. electrostatic.
C. electromagnetism.
D. electromagnetic induction.

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5. An electron travelling horizontally enters a region where a uniform electric field is directed upward as shown in the given figure.



What will be the direction of the force exerted on the electron when it enters into the field?

A. Upward
B. To the left
C. Downward
D. To the right

6. In which of the following media, the force of Coulomb decreases more as compared to the vacuum?

A. Gases
B. Metals
C. Liquids
D. Insulators

7. A student has been tasked to get maximum power dissipation by different dry cells in a laboratory.

To fulfill this task, s/he should keep in mind that the effective internal resistance MUST be

I. equal to the external resistance
II. less than the external resistance
III. more than the external resistance

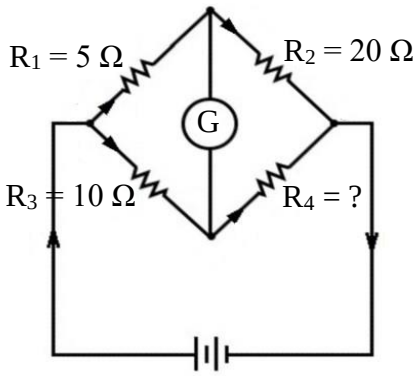
A. I only.
B. III only.
C. I and II.
D. I and III.

8. The terminal potential difference of a battery is equal to its electromotive force (e.m.f.). when its internal resistance become

A. low.
B. zero.
C. high.
D. constant.

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9. The unknown resistance of the given Whetstone bridge is



A. 2.5 Ω.
B. 5 Ω.
C. 20 Ω.
D. 40 Ω.

10. Heat generated by a 40 W bulb in an hour is

A. 40 J.
B. 90 J.
C. 2400 J.
D. 144000 J.

11. If the resistivity and cross-sectional area of a metallic wire remain constant, then on increasing its length the resistance of the wire will

A. increase.
B. decrease.
C. become zero.
D. remain constant.

12. When a shunt is connected parallel with a galvanometre, it is converted into a/ an

A. ammetre.
B. voltmetre.
C. ohmmetre.
D. photometre.

13. If a charged particle of 2 C moves with a velocity of 3 m/s perpendicular to a magnetic field of 1 T, then the magnetic force acting on the particle will be

A. 3 N.
B. 4 N.
C. 5 N.
D. 6 N.

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14. All of the following factors affect the torque acting on a current carrying coil placed in a magnetic field EXCEPT

    A. electric field.
    B. area of the loop.
    C. amount of the current.
    D. direction of the current.

15. All of the following quantities are measured by an AVO metre EXCEPT

    A. current.
    B. voltage.
    C. resistance.
    D. electric flux.

16. If a current carrying conductor is placed at right angle to the magnetic field, then the force experienced by the conductor will

    A. be zero.
    B. be maximum.
    C. remain constant.
    D. vary unpredictably.

17. Which of the following transformers converts 220 V into 9 V?

    A. An ideal transformer
    B. A step-up transformer
    C. A step-down transformer
    D. An automatic transformer

18. Which of the following is the CORRECT effect for the back electromotive force (e.m.f.) in the current of an alternating current (A.C.) motor?

    A. It does not have any effect.
    B. It reverses the direction of current.
    C. It increases the current in the motor.
    D. It decreases the current in the motor.

19. In a transformer, the laminated iron core has reduced the eddy current because the plates in the transformer are

    A. made up of iron metal.
    B. insulated from each other.
    C. separated with metallic laminations.
    D. separated with laminations having winding wires.

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20. Faraday's law of electromagnetic induction is TRUE, when an electromotive force (e.m.f.) is induced in a conductor when it

    A. cuts the magnetic flux.
    B. follows the path of magnetic flux.
    C. moves parallel to the magnetic field.
    D. moves perpendicular to the magnetic field.

21. In an alternating current circuit, the current through a capacitor at a high frequency will be

    A. zero.
    B. large.
    C. small.
    D. undetermined.

22. The waveform of the voltage is sinusoidal and the waveform of the current lags behind it.

    With reference to the given statement, the CORRECT phase relationship between voltage and current is that

    A. they both become zero.
    B. they both become infinite.
    C. the voltage leads the current.
    D. the current leads the voltage.

23. Which of the following is NOT an electromagnetic wave?

    A. Light
    B. Radio
    C. Sound
    D. Infra-red

24. The maximum value that the alternating current or voltage reaches in one cycle is called

    A. peak value.
    B. phase value.
    C. peak to peak value.
    D. root mean square value.

25. Which of the following options does NOT relate to an insulator?

    A. Complete valence band
    B. Partially filled conduction band
    C. Tightly bounded valence electrons
    D. Large energy gapped between electrons

26. If a material becomes more resistant to shear deformation, then shear modulus will

    A. increase.
    B. decrease.
    C. become zero.
    D. remain constant.

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27. A structure that is intermediate between solid and liquid is called

A. metallic solid.
B. polymeric solid.
C. crystalline solid.
D. amorphous solid.

28. The type of impurity that has to be added to a semiconductor material so that it can provide holes is

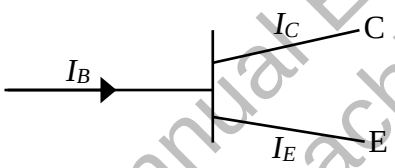
A. trivalent.
B. tetravalent.
C. pentavalent.
D. monovalent.

29. The substance(s) in which the atoms do NOT form magnetic dipole is/ are called

I. ferromagnetic
II. paramagnetic
III. diamagnetic

A. I only.
B. III only.
C. I and II.
D. II and III.

30. The current gained by the given transistor in the common-emitter circuit is defined by the ratio



A. I_C / I_E
B. I_C / I_B
C. I_E / I_B
D. I_E / I_C

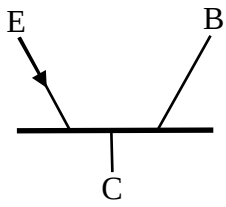
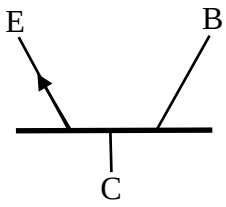
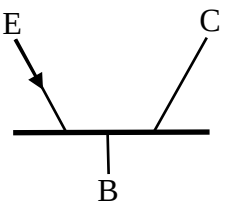
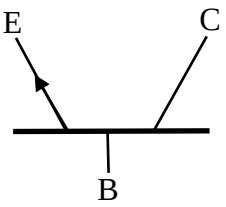
31. In the AND gate truth table, the output become low (1), when

A. one input become low (0).
B. one input become high (1).
C. both inputs become low (0).
D. both inputs become high (1).

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32. The CORRECT symbol of a *n-p-n* transistor is

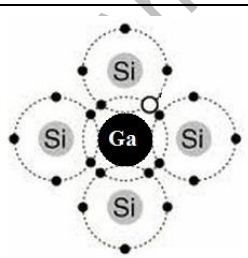
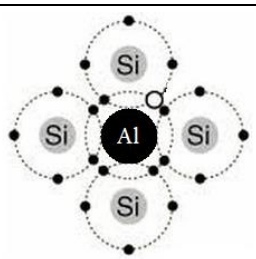
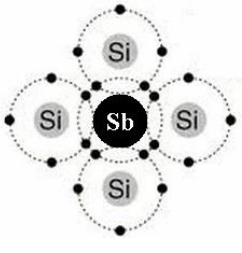
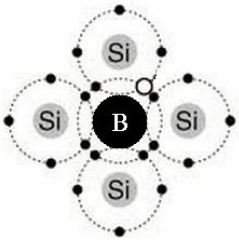
(**Note**: Where 'E', 'B' and 'C' have their usual meaning.)

	
A	B
	
C	D

33. The output of AND gate will be 1 when

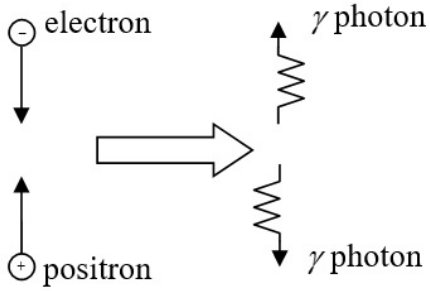
	Input 1	**Input 2**
A	0	0
B	1	1
C	0	1
D	1	0

34. Which of the following is the *n*-type semi-conductor in terms of doping?

	
A	B
	
C	D

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35. Which of the following phenomena is occurring in the given figure?



A. Pair production
B. Pair annihilation
C. Compton's effect
D. Photoelectric effect

36. In the context of the wave nature of light, the diffraction of light

A. changes its direction due to a change in medium.
B. reflects from a smooth surface.
C. bends around the obstacles.
D. refracts through a prism.

37. The statement that is TRUE for the threshold frequency in a photoelectric effect is that it is

A. same for all types of materials.
B. different for same types of materials.
C. less than zero for all types of materials.
D. different for different types of materials.

38. The photoelectric and Compton's effect describe that light possesses

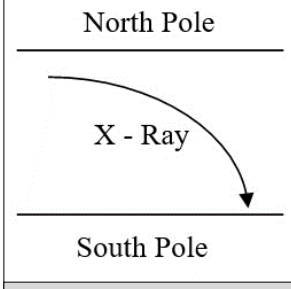
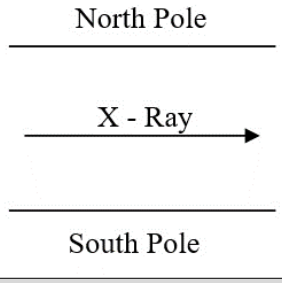
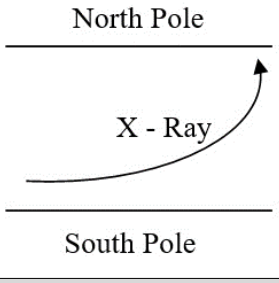
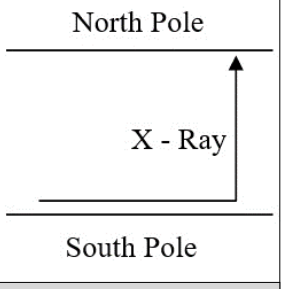
A. dual nature.
B. wave nature only.
C. particle nature only.
D. electromagnetic nature.

39. If the temperature of black body is raised, then the wavelength corresponding to maximum intensity will

A. become zero.
B. remain the same.
C. shift towards longer wavelength.
D. shift towards shorter wavelength.

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40. Which of the following options shows the behaviour of X-rays when it passes through magnetic field?

A	B	C	D
			

41. According to Bohr's atomic model, the emission or absorption of energy by electrons occurs

A. randomly.
B. continuously.
C. when electrons change their orbits.
D. when electrons become positively charged particle.

42. If the transition in hydrogen atom from the excited state (n = 6) to the lower state (p = 3) and Rydberg's constant is $1.0974 \times 10^{7}\ m^{-1}$, then the wavelength of spectral line is

A. 1.093×10^{-6} m.
B. 2.524×10^{-6} m.
C. 3.645×10^{-7} m.
D. 4.140×10^{-7} m.

43. When an electron in an atom jumps from higher energy level to lower energy level, it will

A. emit energy.
B. absorb energy.
C. neither absorb nor emit energy.
D. first absorb and then emit energy.

44. In today's world, food industries use X-rays for

A. sterilisation of food.
B. validation of expiry date.
C. improving cooking process.
D. increasing the shelf life of food.

45. Geiger Muller counter is used to detect

A. energy.
B. particle.
C. electric charge.
D. ionising radiation.

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46. Which of the following statements CORRECTLY describes the charge distribution in an atom?

A. Protons are neutrons orbit around the nucleus.
B. Electrons are evenly distributed in the nucleus.
C. Electrons orbit the nucleus in the specific energy levels.
D. Protons and electrons are evenly distributed in the energy levels.

47. The nucleus $^{82}Pb_{214}$ emits an electron and becomes nuclide **X**.

Which of the following shows the CORRECT atomic and mass number of nuclide **X**?

	Atomic Number	**Mass Number**
A	80	210
B	81	210
C	81	214
D	83	214

48. Which of the following radiations possesses maximum velocity and penetration power?

	Particles with Maximum Velocity	**Particles with Maximum Penetration Power**
A	Alpha	Beta
B	Beta	Alpha
C	Gamma	Gamma
D	Alpha	Alpha

49. The nuclear radiation therapy is suitable for treating

A. cancer.
B. kidney stones.
C. spinal dislocation.
D. nervous breakdown.

50. When an alpha particle encounters an atom, it eliminates

I. proton
II. neutron
III. electron

A. I only.
B. III only.
C. I and II.
D. II and III.

END OF PAPER

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